

SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Deep Learning

COURSE CODE: MLA0403

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

QUESTIONS: 5

Duration: 1 Hour
Total Marks:50

Annexure A

Analytical Problem Solving

1. A coin is tossed 10 times and the outcomes are:

H H T H H H T H H T

Using Maximum Likelihood Estimation, estimate the probability of getting Heads (p).

Solution:

Given Outcomes:

H H T H H H T H H T

Total tosses (n) = 10

Number of Heads (k) = 7

Formula

$$\hat{p} = \frac{k}{n}$$

Calculation

$$\hat{p} = \frac{7}{10} = 0.7$$

Answer

The MLE estimate of the probability of getting Heads is:

$$\hat{p} = 0.7$$

2. Out of 50 students, 42 passed an exam.

Assuming a Bernoulli distribution, find the MLE of the probability that a student passes.

Solution:

Given:

- Total students = 50
- Students passed = 42

Formula

$$\hat{p} = \frac{\text{Number of Passes}}{\text{Total Students}}$$

Calculation

$$\hat{p} = \frac{42}{50} = 0.84$$

Answer

$$\hat{p} = 0.84$$

The estimated probability that a student passes is 0.84 (84%)

3. A factory produces light bulbs. In a sample of 100 bulbs, 8 are defective.

Find the MLE for the probability that a bulb is defective.

Solution:

Given:

- Total bulbs = 100
- Defective bulbs = 8

Formula

$$\hat{p} = \frac{\text{Defective Bulbs}}{\text{Total Bulbs}}$$

Calculation

$$\hat{p} = \frac{8}{100} = 0.08$$

Answer

$$\boxed{\hat{p} = 0.08}$$

The MLE estimate of the defective probability is 0.08 (8%).

4. A biased die is rolled 60 times. The number 6 appears 18 times.

Estimate the probability of rolling a 6 using MLE.

Solution:

Given:

- Total rolls = 60
- Number of sixes = 18

Formula

$$\hat{p} = \frac{\text{Number of Sixes}}{\text{Total Rolls}}$$

Calculation

$$\hat{p} = \frac{18}{60} = 0.3$$

Answer

$$\boxed{\hat{p} = 0.3}$$

The estimated probability of rolling a 6 is 0.3 (30%).

5. Given: Input

$$(x) = 4$$

$$\text{Weight } (w) = 2$$

$$\text{Actual Output} = 10$$

$$\text{Prediction: } y = w \times x$$

1. Find the predicted output.
2. Calculate the error.
3. If the gradient is 4 and learning rate is 0.01, find the updated weight.

Solution:

Given

- Input $(x) = 4$
- Weight $(w) = 2$
- Actual Output = 10
- Gradient = 4
- Learning Rate $(\alpha) = 0.01$

(1) Predicted Output

Formula:

$$y = w \times x$$

Calculation:

$$y = 2 \times 4 = 8$$

Predicted Output

8

(2) Error

Formula:

$$\text{Error} = \text{Actual Output} - \text{Predicted Output}$$

Calculation:

$$\text{Error} = 10 - 8 = 2$$

Error

2

(3) Updated Weight

Formula (Gradient Descent):

$$w_{\text{new}} = w - \alpha \times \text{Gradient}$$

Calculation:

$$w_{\text{new}} = 2 - (0.01 \times 4)$$

$$w_{\text{new}} = 2 - 0.04 = 1.96$$

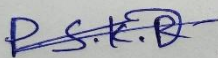
Updated Weight

1.96

Code of Conduct:

I P Surendra Kumar Reddy (192425224) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation